

## Lecture 5

# Characters, the canonical decomposition, and projectors

**Objectives.** The trace compresses each representation matrix to a single number, and these numbers — the characters — turn out to retain everything that matters: they detect equivalence, count multiplicities, and test irreducibility, all through one inner product. We develop the two orthogonality relations, construct character tables for  $C_n$ ,  $S_3$ ,  $D_4$  and  $C_{3v}$  by a systematic algorithm, and convert the theory into operators: the projectors that carve any representation into its symmetry types, the computational workhorse of the applications in Lectures 6 and 7.

## 5.1 Characters and class functions

**Definition 5.1 (Character).** The character of a representation  $(\rho, V)$  of  $G$  is the function

$$\chi_\rho: G \rightarrow \mathbb{C}, \quad \chi_\rho(g) = \text{tr } \rho(g) = \text{tr } D(g),$$

the trace being the same in every basis. The characters of the irreducible representations  $\rho^{(i)}$  are written  $\chi_i$  and called *irreducible characters*. Note  $\chi_\rho(e) = \text{tr } \text{id}_V = \dim V$ : the character remembers the dimension.

**Proposition 5.2 (Basic properties).** Let  $\chi$  be the character of  $(\rho, V)$ , with  $G$  finite. Then:

- (i) equivalent representations have equal characters;
- (ii)  $\chi(hgh^{-1}) = \chi(g)$  for all  $g, h$ : characters are constant on conjugacy classes;
- (iii)  $\chi(g^{-1}) = \overline{\chi(g)}$ ;
- (iv) characters add on direct sums:  $\chi_{\rho \oplus \sigma} = \chi_\rho + \chi_\sigma$ ;
- (v) the dual representation has the conjugate character:  $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$ .

*Proof.* (i) and (ii) are the invariance of the trace under conjugation:  $\text{tr}(SDS^{-1}) = \text{tr } D$  and  $\text{tr}(D(h)D(g)D(h)^{-1}) = \text{tr } D(g)$ . (iii) The element  $g$  has finite order  $m$ , so  $\rho(g)^m = \text{id}$ ; the minimal polynomial of  $\rho(g)$  divides  $x^m - 1$ , which has distinct roots, so  $\rho(g)$  is diagonalisable with eigenvalues  $\lambda_1, \dots, \lambda_n$  that are  $m$ -th roots of unity. Then  $\chi(g) = \sum_a \lambda_a$ , while  $\rho(g^{-1}) = \rho(g)^{-1}$  has eigenvalues  $\lambda_a^{-1} = \overline{\lambda_a}$  (unit modulus), whence  $\chi(g^{-1}) = \overline{\chi(g)}$ . (iv) In a basis adapted to the direct sum the matrices are block diagonal, and the trace adds over blocks. (v) From Lecture 3,  $D^*(g) = D(g^{-1})^T$ , and transposition does not change the trace; then apply (iii). ■

Property (ii) places the characters in a distinguished finite-dimensional space:

**Definition 5.3 (Class functions).** A class function on  $G$  is a function  $G \rightarrow \mathbb{C}$  constant on conjugacy classes. Class functions form a subspace  $\text{CF}(G)$  of the space of all functions on  $G$ ,

of dimension equal to the number of conjugacy classes (one free value per class), which we endow with the inner product of Lecture 4,

$$\langle \varphi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\varphi(g)} \psi(g) = \frac{1}{|G|} \sum_{\text{classes } C} |C| \overline{\varphi(g_C)} \psi(g_C),$$

the second form (with  $g_C$  any representative of  $C$ ) being how one computes in practice.

One more property of characters costs a line now and pays handsomely in Lecture 6:

**Lemma 5.4 (Fixed-point count).** *The character of a permutation representation (Lecture 3) counts fixed points:*

$$\chi_{\text{perm}}(g) = \#\{x \in X : g \cdot x = x\}.$$

*Proof.* The diagonal entries of the permutation matrix are  $D_{xx}(g) = \delta_{x, g \cdot x}$ : a 1 for each fixed point. ■

(In Lecture 6 this becomes the “atoms left in place” rule for the vibrational analysis of molecules.) For the regular representation,  $X = G$  acted on by left translation,  $g \cdot h = gh$  has a fixed point only for  $g = e$ :

$$\chi_{\text{reg}}(e) = |G|, \quad \chi_{\text{reg}}(g) = 0 \quad (g \neq e). \quad (5.1)$$

## 5.2 The orthogonality relations

**Theorem 5.5 (First orthogonality relation).** *The irreducible characters are orthonormal:*

$$\langle \chi_i, \chi_j \rangle_G = \delta_{ij}.$$

*Since the number of irreducible representations equals the number of conjugacy classes (Lecture 4), the irreducible characters form an orthonormal basis of the space of class functions.*

*Proof.* Take unitary matrices and sum the great orthogonality theorem of Lecture 4 over the diagonal:

$$\langle \chi_i, \chi_j \rangle_G = \sum_{\mu=1}^{n_i} \sum_{\alpha=1}^{n_j} \left\langle D_{\mu\mu}^{(i)}, D_{\alpha\alpha}^{(j)} \right\rangle_G = \sum_{\mu, \alpha} \frac{1}{n_i} \delta_{ij} \delta_{\mu\alpha} \delta_{\mu\alpha} = \delta_{ij} \frac{n_i}{n_i} = \delta_{ij}.$$

The characters are thus  $k$  orthonormal — hence linearly independent — vectors in the  $k$ -dimensional space  $\text{CF}(G)$ , where  $k$  is the number of classes: a basis. ■

The same information, read column-wise, gives a second relation that is just as useful in practice:

**Theorem 5.6 (Second orthogonality relation).** *For any  $g, h \in G$ ,*

$$\sum_i \overline{\chi_i(g)} \chi_i(h) = \begin{cases} \frac{|G|}{|[g]|} = |C_G(g)|, & h \text{ conjugate to } g, \\ 0, & \text{otherwise,} \end{cases}$$

*the sum running over all irreducible characters.*

*Proof.* Let  $C_1, \dots, C_k$  be the classes and form the  $k \times k$  matrix

$$X_{iC} = \sqrt{\frac{|C|}{|G|}} \chi_i(g_C).$$

The first orthogonality relation says precisely that the rows of  $X$  are orthonormal,  $XX^\dagger = \mathbb{I}$ ; a square matrix with orthonormal rows is unitary, so its columns are orthonormal too,  $X^\dagger X = \mathbb{I}$ :

$$\frac{\sqrt{|C| |C'|}}{|G|} \sum_i \overline{\chi_i(g_C)} \chi_i(g_{C'}) = \delta_{CC'},$$

which is the assertion (the centraliser form uses  $|[g]| = [G : C_G(g)]$  from Lecture 2). ■

Setting  $g = h = e$  recovers  $\sum_i n_i^2 = |G|$ : the column of the identity carries the dimension theorem of Lecture 4.

### 5.3 Characters know everything

**Theorem 5.7 (Multiplicity formula).** *Let  $(\rho, V)$  be any representation, with character  $\chi_V$  and decomposition  $V \cong \bigoplus_i m_i V^{(i)}$ . Then*

$$m_i = \langle \chi_i, \chi_V \rangle_G :$$

*the multiplicities are inner products, computable from the character alone, with no basis-hunting.*

*Proof.* By additivity (Proposition 5.2(iv)),  $\chi_V = \sum_j m_j \chi_j$ ; take the inner product with  $\chi_i$  and use orthonormality. ■

**Corollary 5.8 (The character is a complete invariant).** *Two representations of a finite group are equivalent if and only if they have the same character.*

*Proof.* Equivalent representations have equal characters (Proposition 5.2(i)). Conversely, equal characters give equal multiplicities by Theorem 5.7, and representations with the same multiplicities are both equivalent to the same direct sum. ■

**Corollary 5.9 (Irreducibility criterion).**  $\langle \chi_V, \chi_V \rangle_G = \sum_i m_i^2$ . *In particular  $V$  is irreducible if and only if  $\langle \chi_V, \chi_V \rangle_G = 1$ .*

*Proof.* Expand  $\chi_V = \sum m_i \chi_i$  in the orthonormal basis and compute the norm. A sum of squares of nonnegative integers equals 1 precisely when one  $m_i$  is 1 and the rest vanish. ■

**Example 5.10 (The  $S_3$  permutation representation, in one line).** Lecture 3 decomposed the permutation representation of  $S_3$  by an explicit change of basis. Watch characters do it without one. By Lemma 5.4, on the classes  $\{e\}$ ,  $\{\text{transpositions}\}$ ,  $\{\text{3-cycles}\}$  (sizes 1, 3, 2) the character is the number of fixed points:

$$\chi_{\text{perm}} = (3, 1, 0).$$

Against the irreducible characters of  $S_3$  (computed in Section 5.4; trivial  $(1, 1, 1)$ , sign  $(1, -1, 1)$ , standard  $(2, 0, -1)$ ):

$$\begin{aligned} m_{\text{triv}} &= \frac{1}{6} [1 \cdot 1 \cdot 3 + 3 \cdot 1 \cdot 1 + 2 \cdot 1 \cdot 0] = 1, \\ m_{\text{sgn}} &= \frac{1}{6} [1 \cdot 1 \cdot 3 + 3 \cdot (-1) \cdot 1 + 2 \cdot 1 \cdot 0] = 0, \\ m_{\text{std}} &= \frac{1}{6} [1 \cdot 2 \cdot 3 + 3 \cdot 0 \cdot 1 + 2 \cdot (-1) \cdot 0] = 1, \end{aligned}$$

recovering  $\mathbb{C}^3 \cong V^{\text{triv}} \oplus V^{\text{std}}$  — including the *absence* of the sign representation, which the explicit method could only discover after the fact. As a cross-check,  $\langle \chi_{\text{perm}}, \chi_{\text{perm}} \rangle_G = \frac{1}{6}(9 + 3 + 0) = 2 = 1^2 + 1^2$ : exactly two irreducible constituents.

**Example 5.11 (The regular representation, again).** With the character (5.1), the multiplicity of  $V^{(i)}$  in the regular representation is

$$m_i = \langle \chi_i, \chi_{\text{reg}} \rangle_G = \frac{1}{|G|} \overline{\chi_i(e)} |G| = n_i :$$

the decomposition theorem of Lecture 4, re-derived in one line. This is the promised payoff of building the character machinery: theorems that required structural arguments become arithmetic.

## 5.4 Character tables

All the character theory of a finite group fits in a single square array: the *character table*, with one row per irreducible character, one column per conjugacy class. Conventionally the first column is the class of the identity — so the first column lists the dimensions  $n_i$  — and the first row is the trivial character, all 1's; class sizes are recorded above the columns, since the inner product needs them. The two orthogonality relations say that the table, suitably weighted, is a unitary matrix: rows are orthonormal with weights  $|C|/|G|$ , columns with weights  $|G|/|C|$  — a complete error-check on every entry.

A systematic construction algorithm, with the tools now in hand:

- (1) find the conjugacy classes and their sizes (Lecture 2);
- (2) the number of irreducible representations equals the number of classes; solve  $\sum_i n_i^2 = |G|$  with  $n_1 = 1$  for the dimension pattern (Lecture 4), which in small groups is usually forced;
- (3) write down the one-dimensional characters: these are homomorphisms  $G \rightarrow \mathbb{C} \setminus \{0\}$ , fixed by consistent assignments to generators compatible with the relations *and with conjugacy* (a one-dimensional character takes equal values on conjugate elements);
- (4) fill the remaining rows from known subrepresentations (traces of explicit matrices) or by orthogonality to the rows already known — column orthogonality against the identity column is especially effective;
- (5) verify the completed table against both orthogonality relations.

**Example 5.12 ( $C_n$ , and  $C_3$  in full).** For the abelian group  $C_n$  every element is its own class, and the  $n$  irreducible characters are the one-dimensional representations  $\chi_k(r^m) = \omega^{km}$ ,

$\omega = e^{2\pi i/n}$  (Lectures 3–4). For  $n = 3$ :

| $C_3$    | $e$ | $r$        | $r^2$      |
|----------|-----|------------|------------|
| $\chi_0$ | 1   | 1          | 1          |
| $\chi_1$ | 1   | $\omega$   | $\omega^2$ |
| $\chi_2$ | 1   | $\omega^2$ | $\omega$   |

Orthogonality of rows is the geometric-series identity  $1 + \omega^k + \omega^{2k} = 0$  for  $k \neq 0$ : character theory on cyclic groups is the arithmetic of roots of unity — the discrete Fourier transform of Lecture 4 in tabular form.

**Example 5.13 ( $S_3$ ).** Classes  $\{e\}$ ,  $\{(1\ 2), (1\ 3), (2\ 3)\}$ ,  $\{(1\ 2\ 3), (1\ 3\ 2)\}$  of sizes 1, 3, 2; dimension pattern (1, 1, 2) from  $1 + 1 + 4 = 6$ . The one-dimensional characters are the trivial and the sign. The two-dimensional row can be read off from the standard-representation matrices of Lecture 3,  $\text{tr } B((1\ 2)) = 0$  and  $\text{tr } B((1\ 2\ 3)) = -1$  — or, knowing no matrices at all, from column orthogonality against the identity column,  $\sum_i \overline{\chi_i(e)} \chi_i(c) = 0$  for each class  $c \neq \{e\}$ : for the transpositions,  $1 \cdot 1 + 1 \cdot (-1) + 2 \chi_{\text{std}}((1\ 2)) = 0$  gives  $\chi_{\text{std}}((1\ 2)) = 0$ ; for the 3-cycles,  $1 \cdot 1 + 1 \cdot 1 + 2 \chi_{\text{std}}((1\ 2\ 3)) = 0$  gives  $\chi_{\text{std}}((1\ 2\ 3)) = -1$ . The table:

| $S_3$                | 1 $e$ | 3 (1 2) | 2 (1 2 3) |
|----------------------|-------|---------|-----------|
| $\chi_{\text{triv}}$ | 1     | 1       | 1         |
| $\chi_{\text{sgn}}$  | 1     | -1      | 1         |
| $\chi_{\text{std}}$  | 2     | 0       | -1        |

(class sizes prefixed to the representatives). Check:  $\langle \chi_{\text{std}}, \chi_{\text{std}} \rangle_G = \frac{1}{6}(4 + 0 + 2) = 1$  — irreducible, as proved by hand in Lecture 3.

**Example 5.14 ( $D_4$ ).** Classes (Lecture 2):  $\{e\}$ ,  $\{r^2\}$ ,  $\{r, r^3\}$ ,  $\{s, r^2s\}$ ,  $\{rs, r^3s\}$ , sizes 1, 1, 2, 2, 2; dimension pattern (1, 1, 1, 1, 2), forced (Lecture 4). For the one-dimensional characters  $\nu$ : conjugacy of  $r$  with  $r^{-1} = r^3$  forces  $\nu(r) = \nu(r)^{-1}$ , i.e.  $\nu(r) = \pm 1$ , and  $\nu(s) = \pm 1$  from  $s^2 = e$ ; all four sign choices are consistent with the relations, and  $\nu(r^2) = \nu(r)^2 = 1$  always. The two-dimensional row follows from column orthogonality against the identity column: for each class  $c \neq \{e\}$ ,

$$\sum_i \overline{\chi_i(e)} \chi_i(c) = 0 \implies \chi_5(c) = -\frac{1}{2} \sum_{1\text{-dim}} \nu(c),$$

giving  $\chi_5(r^2) = -2$  and  $\chi_5(r) = \chi_5(s) = \chi_5(rs) = 0$ :

| $D_4$    | 1 $e$ | 1 $r^2$ | 2 $r$ | 2 $s$ | 2 $rs$ |
|----------|-------|---------|-------|-------|--------|
| $\chi_1$ | 1     | 1       | 1     | 1     | 1      |
| $\chi_2$ | 1     | 1       | 1     | -1    | -1     |
| $\chi_3$ | 1     | 1       | -1    | 1     | -1     |
| $\chi_4$ | 1     | 1       | -1    | -1    | 1      |
| $\chi_5$ | 2     | -2      | 0     | 0     | 0      |

As a check, the defining representation  $D(r) = R(\pi/2)$ ,  $D(s) = \text{diag}(-1, 1)$  has traces 2, -2, 0, 0, 0 — it is  $\chi_5$ , and  $\langle \chi_5, \chi_5 \rangle_G = \frac{1}{8}(4 + 4 + 0 + 0 + 0) = 1$  confirms irreducibility. The quaternion group  $Q_8$ , despite being non-isomorphic to  $D_4$ , has an *identical* character table (Sheet 5): the character table is a powerful invariant, but not a complete one for the group itself.

**Example 5.15** ( $C_{3v}$ , in the chemists' notation). The point group  $C_{3v}$  of ammonia is isomorphic to  $S_3$  (Lecture 1), so its character table contains the same numbers; but the applications of Lecture 6 will use the notation in which chemists and spectroscopists tabulate it. Classes are labelled by operation type — the identity  $E$ , the two rotations  $2C_3$ , the three vertical reflections  $3\sigma_v$  — and the irreducible representations carry *Mulliken symbols*:

| $C_{3v}$ | $E$ | $2C_3$ | $3\sigma_v$ |
|----------|-----|--------|-------------|
| $A_1$    | 1   | 1      | 1           |
| $A_2$    | 1   | 1      | -1          |
| $E$      | 2   | -1     | 0           |

The conventions, in the part we need:  $A$  and  $B$  denote one-dimensional representations, symmetric ( $\chi(C_n) = +1$ ) respectively antisymmetric ( $-1$ ) under the principal rotation; the subscript 1 or 2 records symmetry or antisymmetry under the vertical reflections;  $E$  denotes a two-dimensional irreducible and  $T$  a three-dimensional one. Beware the standard double duty of the letter  $E$ : the identity *operation* (a column) and the two-dimensional *representation* (a row); context always disambiguates. Likewise these notes write  $e$  for the abstract identity element but  $E$  in Schoenflies tables — both usages are entrenched, and we keep each in its habitat.

## 5.5 The canonical decomposition and projection operators

Characters compute multiplicities; we now make them compute *subspaces*. Two notational conventions, used from here through Lecture 7: superscripts in parentheses,  $V^{(i)}$ , always denote the abstract irreducible representations; subscripts,  $V_i$ , denote the pieces of a given representation  $V$  defined next.

**Definition 5.16** (Isotypic components). Let  $(\rho, V)$  be a representation. The *isotypic component*  $V_i$  of type  $i$  is the sum of all irreducible invariant subspaces of  $V$  isomorphic to  $V^{(i)}$ .

If  $V \cong \bigoplus_i m_i V^{(i)}$ , the component  $V_i$  collects the  $m_i$  copies of  $V^{(i)}$ , so  $\dim V_i = m_i n_i$ . The individual copies inside  $V_i$  are *not* canonical when  $m_i > 1$  (Lecture 3's caveat), but the components themselves are, as the following theorem shows by exhibiting each  $V_i$  as the image of an operator built solely from the character.

**Theorem 5.17** (Canonical decomposition and projectors). For each irreducible representation define

$$P_i = \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g).$$

Then:

- (i) each  $P_i$  commutes with every  $\rho(h)$  and preserves every invariant subspace;
- (ii) on an irreducible invariant subspace  $W \subseteq V$  of type  $j$ ,  $P_i$  acts as  $\delta_{ij} \text{id}_W$ ;
- (iii)  $P_i^2 = P_i$ ,  $P_i P_j = 0$  for  $i \neq j$ , and  $\sum_i P_i = \text{id}_V$ ;
- (iv)  $P_i$  is the projection onto the isotypic component  $V_i$  along the sum of the others:

$$V = \bigoplus_i V_i, \quad V_i = \text{im } P_i,$$

a decomposition canonically attached to  $V$ ;

$$(v) \quad \text{tr } P_i = m_i n_i = \dim V_i.$$

*Proof.* (i) Conjugating and substituting  $g' = hgh^{-1}$ ,

$$\rho(h)P_i\rho(h)^{-1} = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \rho(hgh^{-1}) = \frac{n_i}{|G|} \sum_{g'} \overline{\chi_i(h^{-1}g'h)} \rho(g') = P_i,$$

because  $\chi_i$  is a class function. If  $W$  is invariant then  $\rho(g)W \subseteq W$  for every  $g$ , hence  $P_i W \subseteq W$ .

(ii) By (i), the restriction  $P_i|_W$  commutes with the irreducible action of  $G$  on  $W$ , so Schur's lemma gives  $P_i|_W = \lambda \text{id}_W$ . The trace determines  $\lambda$ :

$$\lambda n_j = \text{tr}(P_i|_W) = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \chi_j(g) = n_i \langle \chi_i, \chi_j \rangle_G = n_i \delta_{ij},$$

so  $\lambda = \delta_{ij}$ .

(iii)–(iv) Decompose  $V$  into irreducible invariant subspaces (Maschke). By (ii),  $P_i$  restricts to the identity on each type- $i$  summand and to zero on every other summand; all the algebraic relations in (iii) follow by checking them summand by summand, and the image of  $P_i$  is exactly the span of the type- $i$  summands. That span contains every irreducible subspace  $U \cong V^{(i)}$  of  $V$  whatsoever (again by (ii),  $P_i|_U = \text{id}$  gives  $U \subseteq \text{im } P_i$ ), so  $\text{im } P_i$  is the isotypic component of Definition 5.16 — and the formula for  $P_i$  makes no reference to the chosen decomposition, so  $V_i$  is canonical.

$$(v) \quad \text{tr } P_i = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \chi_V(g) = n_i \langle \chi_i, \chi_V \rangle_G = n_i m_i \text{ by the multiplicity formula.} \quad \blacksquare$$

The identity  $\sum_i P_i = \text{id}_V$  has a pleasant direct verification: by the second orthogonality relation (column of the identity),  $\sum_i n_i \overline{\chi_i(g)} = |G| \delta_{g,e}$ , so the sum collapses to  $\rho(e)$ .

**Example 5.18 (The three projectors of the  $S_3$  permutation representation).** Take  $V = \mathbb{C}^3$  with the permutation matrices of Lecture 3, and let  $J$  denote the  $3 \times 3$  matrix of ones. Summing the six permutation matrices puts a 2 in every entry (for fixed  $(i, j)$ , exactly two permutations send  $j$  to  $i$ ), so

$$P_{\text{triv}} = \frac{1}{6} \sum_g \rho(g) = \frac{1}{3} J,$$

the rank-one projection onto the diagonal  $\mathbb{C}(1, 1, 1)$  — precisely the “average over the group” operator. For the standard component, with  $C = D((1\ 2\ 3))$  and using  $\mathbb{K} + C + C^2 = J$  (the three cyclic matrices have disjoint supports filling  $J$ ):

$$P_{\text{std}} = \frac{2}{6} [2\mathbb{K} - C - C^2] = \frac{1}{3} (3\mathbb{K} - J) = \mathbb{K} - \frac{1}{3} J,$$

the projection onto the plane  $\sum_i v_i = 0$ . And the three transposition matrices likewise sum to  $J$  (each entry of  $J$  hit exactly once), so

$$P_{\text{sgn}} = \frac{1}{6} [\mathbb{K} + C + C^2 - (\text{sum of transposition matrices})] = \frac{1}{6} (J - J) = 0 :$$

the projector detects the absence of the sign representation by vanishing identically. The checks of Theorem 5.17 are immediate:  $P_{\text{triv}} + P_{\text{std}} + P_{\text{sgn}} = \mathbb{K}$ , the traces are 1, 2, 0 =  $m_i n_i$ , and each operator squares to itself.

### 5.5.1 Finer projectors: partner functions

The character projector  $P_i$  lands in the isotypic component but does not see inside it. When the matrices  $D^{(i)}$  themselves are known (not just their traces), one can resolve  $V_i$  direction by direction. Fix unitary matrices  $D^{(i)}$  and define, for  $1 \leq \mu, \nu \leq n_i$ ,

$$P_{\mu\nu}^{(i)} = \frac{n_i}{|G|} \sum_{g \in G} \overline{D_{\mu\nu}^{(i)}(g)} \rho(g).$$

Call a basis  $(w_1, \dots, w_{n_i})$  of an irreducible subspace  $W \cong V^{(i)}$  *adapted* (its members *partner functions*) if the action takes the standard matrix form  $\rho(g)w_\nu = \sum_\mu D_{\mu\nu}^{(i)}(g) w_\mu$ .

**Theorem 5.19 (Matrix-element projectors).** (i)  $P_{\mu\nu}^{(i)} P_{\alpha\beta}^{(j)} = \delta_{ij} \delta_{\nu\alpha} P_{\mu\beta}^{(i)}$ ; in particular the diagonal operators  $P_{\mu\mu}^{(i)}$  are commuting projections, mutually orthogonal across all  $(i, \mu)$ ;

(ii)  $\sum_{\mu=1}^{n_i} P_{\mu\mu}^{(i)} = P_i$ ;

(iii) on an adapted basis of any irreducible subspace of type  $i$ ,

$$P_{\mu\nu}^{(i)} w_\beta = \delta_{\nu\beta} w_\mu,$$

while every  $P_{\mu\nu}^{(i)}$  annihilates irreducible subspaces of type  $j \neq i$ . Thus  $P_{\mu\mu}^{(i)}$  projects onto the “ $\mu$ -th direction” of each type- $i$  copy, and  $P_{\mu\nu}^{(i)}$  transfers the  $\nu$ -th partner into the  $\mu$ -th: a ladder between partner functions.

*Proof.* (iii) first, since it is the cleanest:

$$P_{\mu\nu}^{(i)} w_\beta = \frac{n_i}{|G|} \sum_g \overline{D_{\mu\nu}^{(i)}(g)} \sum_\alpha D_{\alpha\beta}^{(j)}(g) w_\alpha = \sum_\alpha \left[ n_i \left\langle D_{\mu\nu}^{(i)}, D_{\alpha\beta}^{(j)} \right\rangle_G \right] w_\alpha = \delta_{ij} \delta_{\nu\beta} w_\mu$$

by the great orthogonality theorem. (ii) follows by summing the definition over  $\mu = \nu$ , since  $\sum_\mu D_{\mu\mu}^{(i)} = \chi_i$ . For (i), evaluate both sides on an adapted basis of each irreducible summand of  $V$  (Maschke): by (iii), the left side sends  $w_\beta$  (type  $k$ ) to  $\delta_{jk} \delta_{\alpha\beta} P_{\mu\nu}^{(i)} w_\alpha = \delta_{jk} \delta_{\alpha\beta} \delta_{ik} \delta_{\nu\alpha} w_\mu$ , and the right side to  $\delta_{ij} \delta_{\nu\alpha} \delta_{ik} \delta_{\alpha\beta} w_\mu$  — equal. Two operators agreeing on a basis of  $V$  are equal. ■

Unlike  $P_i$ , the finer projectors depend on the chosen matrix realisation  $D^{(i)}$  — rotate the basis of  $V^{(i)}$  and the  $P_{\mu\nu}^{(i)}$  rotate accordingly; only their sum  $P_i$  is canonical. In applications this is a feature, not a bug: one chooses the  $D^{(i)}$  adapted to the physics (say, aligned with molecular axes), and then uses enough independent starting vectors to span every multiplicity space. Fix a reference index  $\nu$  for type  $i$ , apply  $P_{\nu\nu}^{(i)}$  to candidate vectors, discard zero results, and retain (or orthonormalise) independent results until they span  $\text{im } P_{\nu\nu}^{(i)}$ . For each retained vector  $u$ , the transfer operators  $P_{\mu\nu}^{(i)}$  manufacture its partners  $P_{\mu\nu}^{(i)} u$  for all  $\mu$ . Repeat for every type  $i$ ; the resulting vectors span the whole representation. A single seed can miss an isotypic component or fail to resolve repeated copies, whereas this rank-aware procedure cannot. Lecture 6 practises the coarser, canonical character/isotypic-projector method on the normal modes of  $\text{H}_2\text{O}$  and  $\text{NH}_3$ ; it does not carry out this finer rank-aware construction. Lecture 7 then uses the resulting isotypic organisation for degenerate quantum levels.

**Remark 5.20 (Computer algebra: the whole lecture as a notebook).** Everything above mechanises. The accompanying Mathematica session (i) recomputes conjugacy classes by the

brute-force gather of Lecture 2; (ii) assembles candidate character tables and verifies both orthogonality relations as matrix identities,  $XX^\dagger = X^\dagger X = \mathbb{I}$  for the weighted table  $X$ ; (iii) computes characters of given representations as traces, multiplicities as the inner products of Theorem 5.7, and irreducibility via  $\langle \chi, \chi \rangle_G$ ; and (iv) builds the projectors  $P_i$  and  $P_{\mu\nu}^{(i)}$  as explicit matrices, confirming Example 5.18 —  $P_{\text{triv}} = J/3$ ,  $P_{\text{std}} = \mathbb{I} - J/3$ ,  $P_{\text{sgn}} = 0$  — together with the rank and idempotency checks. Exact arithmetic with root-of-unity entries (`Exp[2 Pi I k/n]`, `FullSimplify`) keeps every verification symbolic.

### Summary.

- *Character.*  $\chi(g) = \text{tr } \rho(g)$ , a class function invariant under equivalence (Definition 5.1, Proposition 5.2).
- *Orthogonality.* Irreducible characters are orthonormal (first relation, Theorem 5.5); their columns obey a dual relation (Theorem 5.6).
- *Multiplicities.*  $\langle \chi, \chi_i \rangle$  is the multiplicity of the  $i$ th irreducible (Theorem 5.7);  $\|\chi\|^2 = 1$  iff the representation is irreducible (Corollary 5.9).
- *Complete invariant.* The character determines a representation up to equivalence (Corollary 5.8).
- *Projectors.* Character projectors give the canonical decomposition into isotypic components (Theorem 5.17).

**Tutorial.** Exercise Sheet 5.

### References

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